

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical
Practical – I

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Class: TYIT	Batch: 01
Date of Assignment: 24/07/2026	Date/Time of Submission:24/07//2026

1. Introduction to Business Analytics:

a. Collect data from a real-life business scenario and perform exploratory data analysis (EDA) to gain insights into the dataset.

```
#Import Required Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
#Load dataset
df = pd.read_csv("Retail_sales.csv")

#Display First 5 Records
df.head()
```

	Store ID	Product ID	Date	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Store Location	Product Category	Day of the Week	Holiday Effect
0	Spearsland	52372247	2022-01-01	9	2741.69	20	81	Tanzania	Furniture	Saturday	False
1	Spearsland	52372247	2022-01-02	7	2665.53	0	0	Mauritania	Furniture	Sunday	False
2	Spearsland	52372247	2022-01-03	1	380.79	0	0	Saint Pierre and Miquelon	Furniture	Monday	False
3	Spearsland	52372247	2022-01-04	4	1523.16	0	0	Australia	Furniture	Tuesday	False
4	Spearsland	52372247	2022-01-05	2	761.58	0	0	Swaziland	Furniture	Wednesday	False

```
#Display Last 5 Records
df.tail()
```

	Store ID	Product ID	Date	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Store Location	Product Category	Day of the Week	Holiday Effect
29995	Spearsland	50239115	2022-01-25	5	2501.15	0	100	Malawi	Clothing	Tuesday	0
29996	Spearsland	50239115	2022-01-26	3	1500.69	0	0	Sudan	Clothing	Wednesday	0
29997	Spearsland	50239115	2022-01-27	6	3001.38	0	0	South Georgia and the South Sandwich Islands	Clothing	Thursday	0
29998	Spearsland	50239115	2022-01-28	5	2501.15	0	0	Haiti	Clothing	Friday	0
29999	Spearsland	50239115	2022-01-29	3	1425.66	5	190	Mozambique	Clothing	Saturday	0

```
#Display Shape of Dataset
```

```
df.shape
```

```
(30000, 11)
```

```
#Display Column Names
```

```
df.columns
```

```
Index(['Store ID', 'Product ID', 'Date', 'Units Sold', 'Sales Revenue (USD)',  
      'Discount Percentage', 'Marketing Spend (USD)', 'Store Location',  
      'Product Category', 'Day of the Week', 'Holiday Effect'],  
      dtype='object')
```

```
#Display Dataset Information
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 30000 entries, 0 to 29999  
Data columns (total 11 columns):  
#   Column                Non-Null Count  Dtype  
---  ----  
0   Store ID              30000 non-null  object  
1   Product ID            30000 non-null  int64  
2   Date                  30000 non-null  object  
3   Units Sold            30000 non-null  int64  
4   Sales Revenue (USD)   30000 non-null  float64  
5   Discount Percentage    30000 non-null  int64  
6   Marketing Spend (USD)  30000 non-null  int64  
7   Store Location         30000 non-null  object  
8   Product Category      30000 non-null  object  
9   Day of the Week       30000 non-null  object  
10  Holiday Effect         30000 non-null  bool  
dtypes: bool(1), float64(1), int64(4), object(5)  
memory usage: 2.3+ MB
```

```
#Display Data Types
```

```
df.dtypes
```

```
Store ID                object  
Product ID              int64  
Date                    object  
Units Sold              int64  
Sales Revenue (USD)     float64  
Discount Percentage     int64  
Marketing Spend (USD)   int64  
Store Location           object  
Product Category        object  
Day of the Week         object  
Holiday Effect          bool  
dtype: object
```

```
#Check Missing Values
df.isnull().sum()
```

Store ID	0
Product ID	0
Date	0
Units Sold	0
Sales Revenue (USD)	0
Discount Percentage	0
Marketing Spend (USD)	0
Store Location	0
Product Category	0
Day of the Week	0
Holiday Effect	0
dtype: int64	

```
#Check Duplicate Records
df.duplicated().sum()
```

np.int64(0)

```
#Display Descriptive Statistics
df.describe()
```

	Product ID	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)
count	3.000000e+04	30000.000000	30000.000000	30000.000000	30000.000000
mean	4.461294e+07	6.161967	2749.509593	2.973833	49.944033
std	2.779759e+07	3.323929	2568.639288	5.974530	64.401655
min	3.636541e+06	0.000000	0.000000	0.000000	0.000000
25%	2.228600e+07	4.000000	882.592500	0.000000	0.000000
50%	4.002449e+07	6.000000	1902.420000	0.000000	1.000000
75%	6.559352e+07	8.000000	3863.920000	0.000000	100.000000
max	9.628253e+07	56.000000	27165.880000	20.000000	199.000000

```
#Count Unique Values
df.nunique()
```

Store ID	1
Product ID	42
Date	731
Units Sold	32
Sales Revenue (USD)	2545
Discount Percentage	5
Marketing Spend (USD)	200
Store Location	243
Product Category	4
Day of the Week	7
Holiday Effect	2
dtype: int64	

```
#Display Unique Values of a Particular Column
df["Product Category"].unique()
```

```
array(['Furniture', 'Electronics', 'Groceries', 'Clothing'], dtype=object)
```

```
#Count Product Categories  
df["Product Category"].value_counts()
```

```
Product Category  
Furniture      9503  
Electronics    8041  
Clothing       6608  
Groceries      5848  
Name: count, dtype: int64
```

```
#Count Store Locations  
df["Store Location"].value_counts()
```

```
Store Location  
Korea          237  
Congo          234  
Turks and Caicos Islands  156  
Guam           153  
Tanzania       151  
...  
Trinidad and Tobago  98  
Haiti          96  
Dominica       95  
Ireland        92  
Andorra        80  
Name: count, Length: 243, dtype: int64
```

```
#Count Holiday Effect  
df["Holiday Effect"].value_counts()
```

```
Holiday Effect  
False    29836  
True      164  
Name: count, dtype: int64
```

```
#Calculate Average Sales Revenue  
df["Sales Revenue (USD)"].mean()
```

```
np.float64(2749.509592666667)
```

```
#Find Maximum Sales Revenue  
df["Sales Revenue (USD)"].max()
```

```
np.float64(27165.88)
```

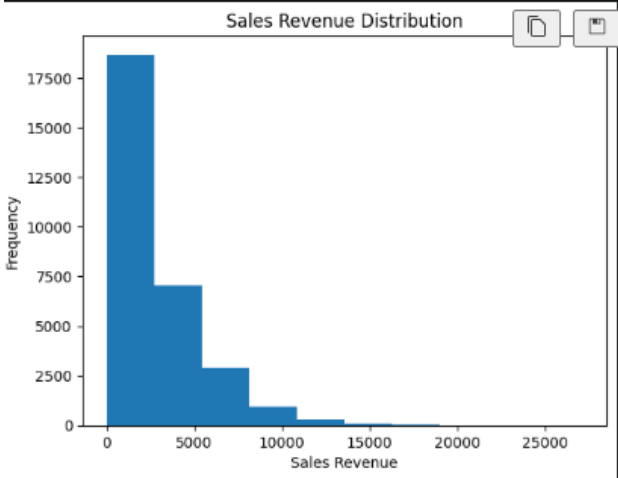
```
#Find Minimum Sales Revenue  
df["Sales Revenue (USD)"].min()
```

```
np.float64(0.0)
```

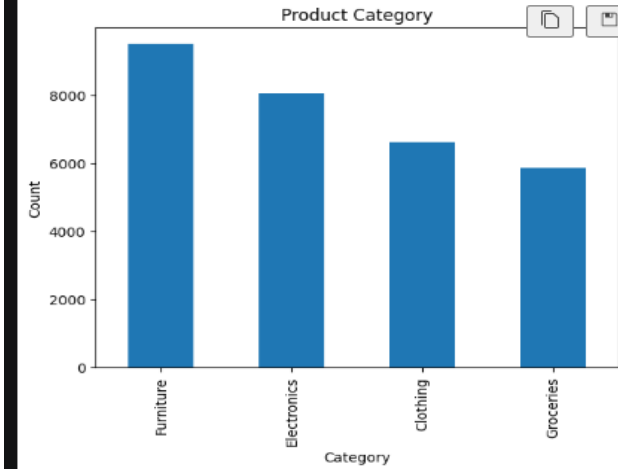
```
#Correlation Matrix  
df.corr(numeric_only=True)
```

	Product ID	Units Sold	Sales Revenue (USD)	Discount Percentage	Marketing Spend (USD)	Holiday Effect
Product ID	1.000000	-0.008570	-0.181934	-0.006891	-0.002496	-0.000015
Units Sold	-0.008570	1.000000	0.615767	-0.001620	-0.004750	0.159324
Sales Revenue (USD)	-0.181934	0.615767	1.000000	-0.065791	-0.002732	0.090681
Discount Percentage	-0.006891	-0.001620	-0.065791	1.000000	-0.000322	0.005848
Marketing Spend (USD)	-0.002496	-0.004750	-0.002732	-0.000322	1.000000	-0.000455
Holiday Effect	-0.000015	0.159324	0.090681	0.005848	-0.000455	1.000000

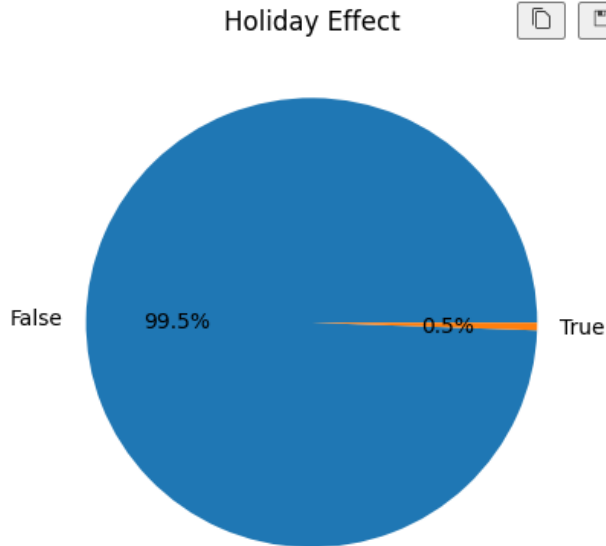
```
#Histogram
plt.hist(df["Sales Revenue (USD)"])
plt.title("Sales Revenue Distribution")
plt.xlabel("Sales Revenue")
plt.ylabel("Frequency")
plt.show()
```



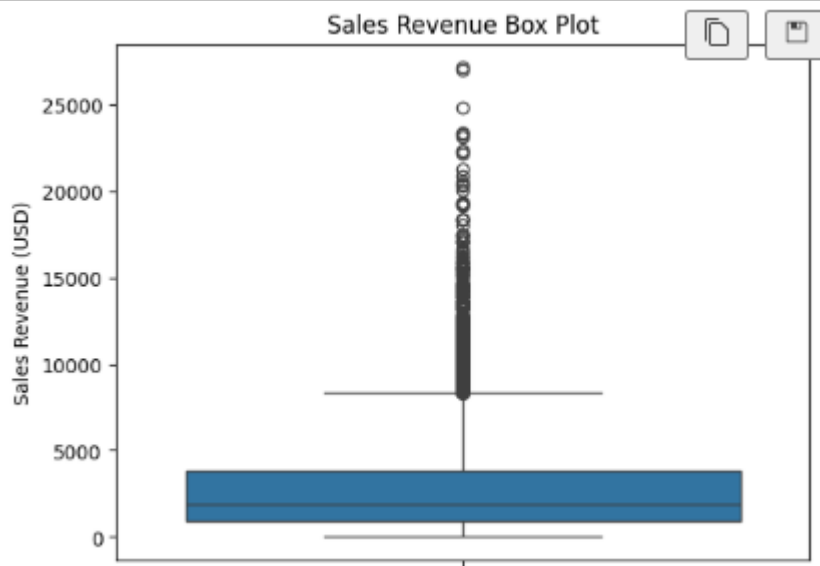
```
#Bar chart
df["Product Category"].value_counts().plot(kind="bar")
plt.title("Product Category")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()
```



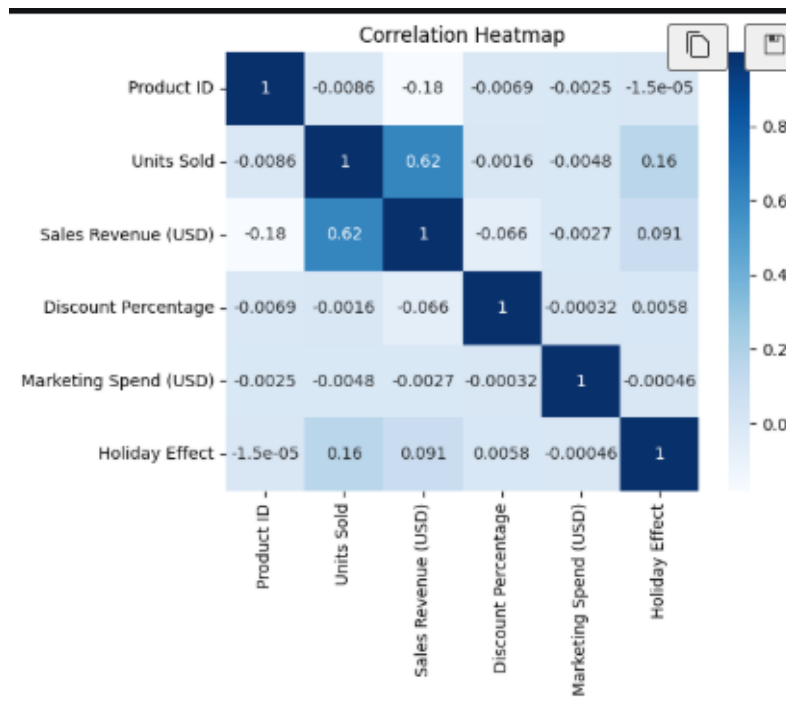
```
#Pie chart
df["Holiday Effect"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Holiday Effect")
plt.ylabel("")
plt.show()
```



```
#Box plot
sns.boxplot(y=df["Sales Revenue (USD)"])
plt.title("Sales Revenue Box Plot")
plt.show()
```



```
#Correlation Heatmap
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Blues")
plt.title("Correlation Heatmap")
plt.show()
```



b. Analyze customer data to identify trends and patterns that can be used for business decision-making.

```
#Import Required Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

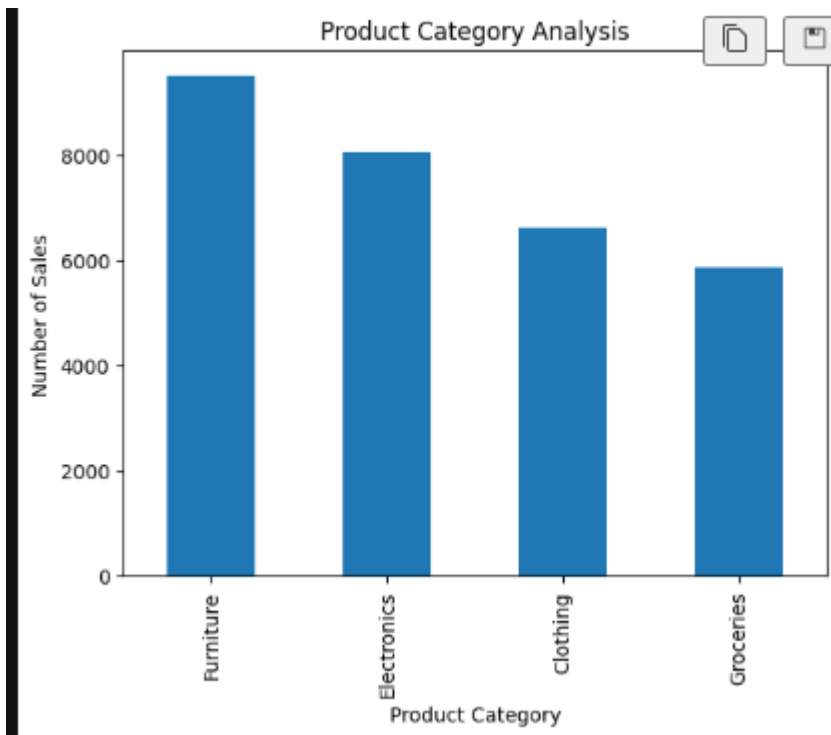
```
#Load dataset
df = pd.read_csv("Retail_sales.csv")
```

Analysis 1 - Product category analysis

```
print(df["Product Category"].value_counts())
```

```
Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64
```

```
df["Product Category"].value_counts().plot(kind="bar")
plt.title("Product Category Analysis")
plt.xlabel("Product Category")
plt.ylabel("Number of Sales")
plt.show()
```

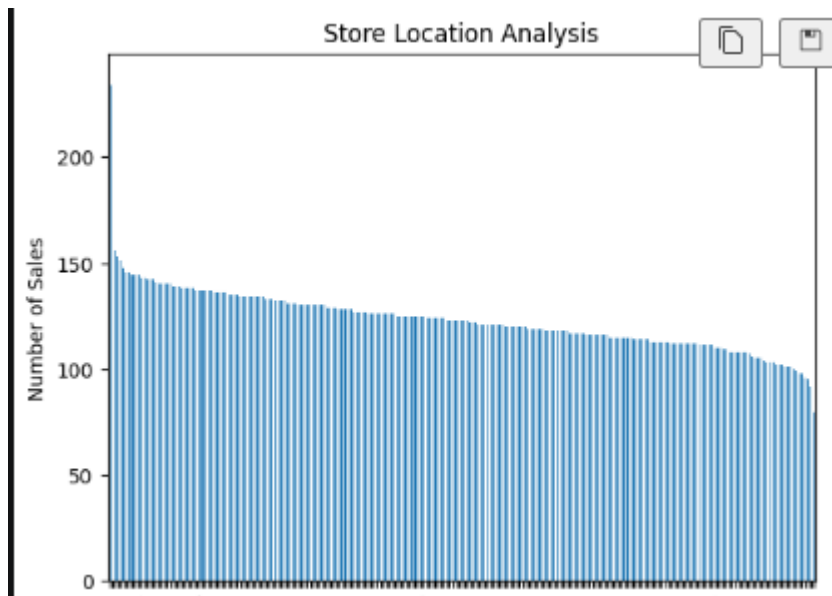


Analysis 2 - Store Location Analysis

```
print(df["Store Location"].value_counts())
```

```
Store Location
Korea                237
Congo                234
Turks and Caicos Islands  156
Guam                 153
Tanzania             151
...
Trinidad and Tobago   98
Haiti                 96
Dominica              95
Ireland               92
Andorra               80
Name: count, Length: 243, dtype: int64
```

```
df["Store Location"].value_counts().plot(kind="bar")
plt.title("Store Location Analysis")
plt.xlabel("Store Location")
plt.ylabel("Number of Sales")
plt.show()
```

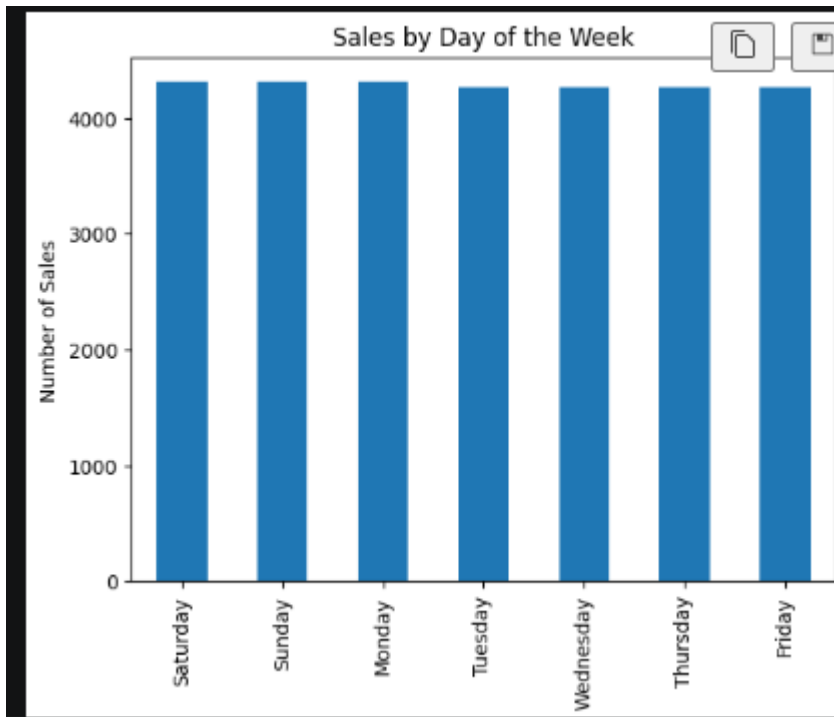


Analysis 3 - Day of the Week Analysis

```
print(df["Day of the Week"].value_counts())
```

```
Day of the Week
Saturday      4310
Sunday        4309
Monday        4309
Tuesday       4268
Wednesday     4268
Thursday      4268
Friday        4268
Name: count, dtype: int64
```

```
df["Day of the Week"].value_counts().plot(kind="bar")
plt.title("Sales by Day of the Week")
plt.xlabel("Day")
plt.ylabel("Number of Sales")
plt.show()
```

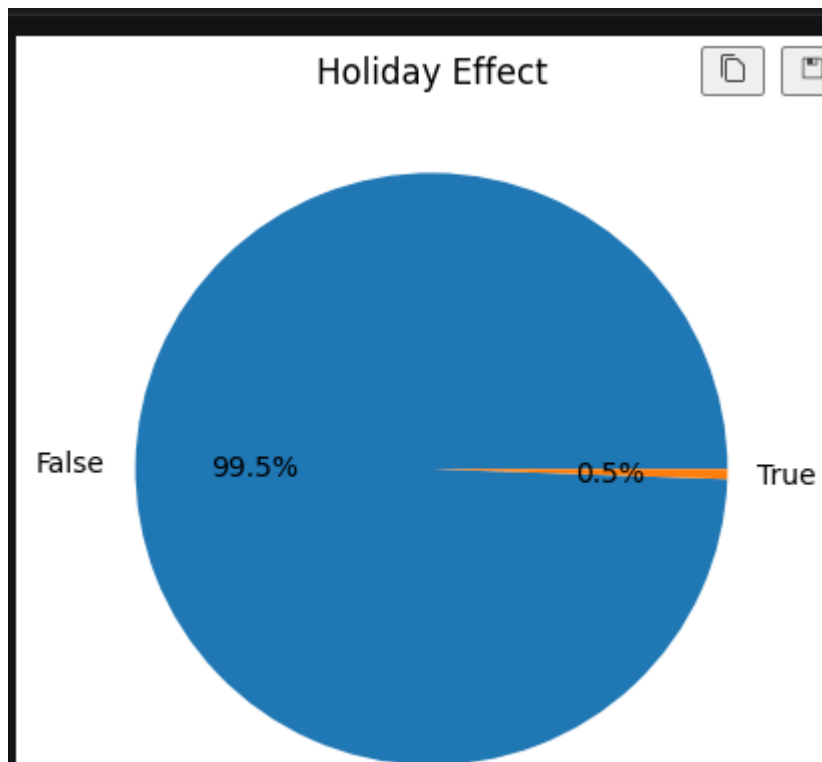


Analysis 4 - Holiday Effect Analysis

```
print(df["Holiday Effect"].value_counts())
```

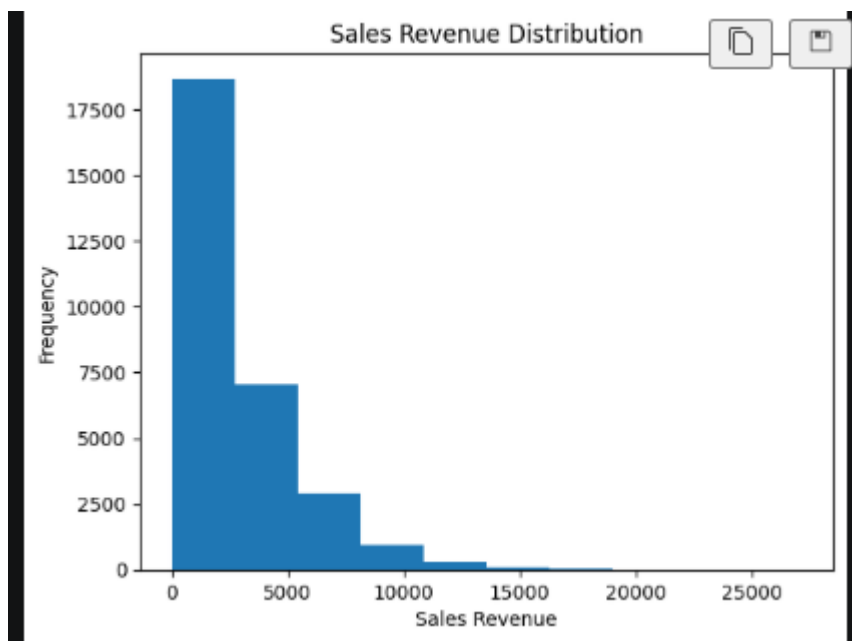
```
Holiday Effect
False      29836
True        164
Name: count, dtype: int64
```

```
df["Holiday Effect"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Holiday Effect")
plt.ylabel("")
plt.show()
```



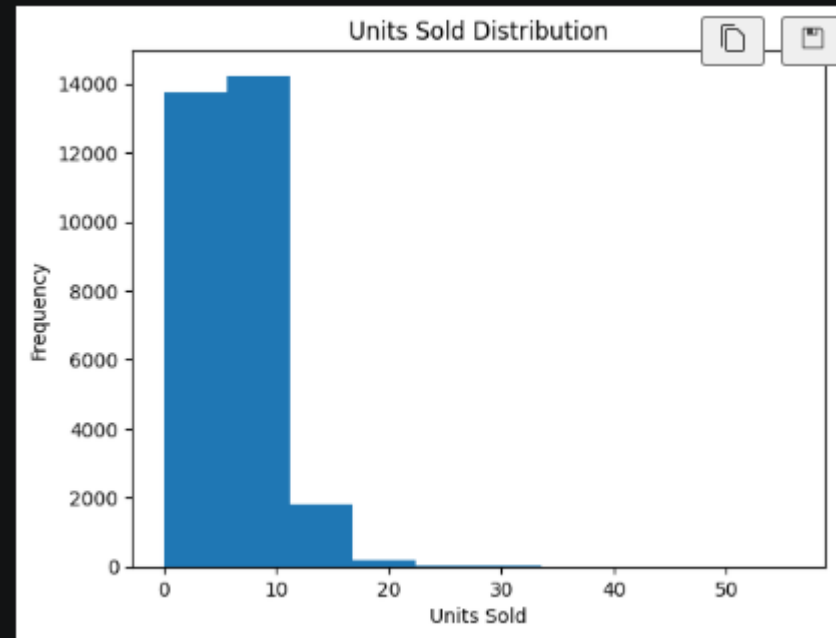
Analysis 5 - Sales Revenue Analysis

```
plt.hist(df["Sales Revenue (USD)"], bins=10)
plt.title("Sales Revenue Distribution")
plt.xlabel("Sales Revenue")
plt.ylabel("Frequency")
plt.show()
```



Analysis 6 - Units Sold Analysis

```
plt.hist(df["Units Sold"], bins=10)
plt.title("Units Sold Distribution")
plt.xlabel("Units Sold")
plt.ylabel("Frequency")
plt.show()
```



Analysis 7 - Marketing Spend vs Sales Revenue

```
plt.scatter(df["Marketing Spend (USD)"], df["Sales Revenue (USD)"])
plt.title("Marketing Spend vs Sales Revenue")
plt.xlabel("Marketing Spend")
plt.ylabel("Sales Revenue")
plt.show()
```



Analysis 8 - Correlation Analysis

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Oranges")
plt.title("Correlation Heatmap")
plt.show()
```

